Reason: Desired genes are inserted without introducing the undesriable genes in target organisms.

VI Answer in a sentence

1. Give the name of wheat variety having higher dietary fibre and protein.
2. Semi-dwarf varieties were introduced in rice. This was made possible by the presence of dwarfing gene in rice. Name this dwarfing gene.
3. Define genetic engineering.
4. Name the types of stem cells.
5. What are transgenic organisms?
6. State the importance of biofertiliser.

VII Short answers questions

1. Discuss the method of breeding for disease resistance.
2. Name three improved characteristics of wheat that helped India to achieve high productivity.
3. Name two maize hybrids rich in amino acid lysine
4. Distinguish between
   a. somatic gene therapy and germ line gene therapy
   b. undifferentiated cells and differentiated cells
5. State the applications of DNA fingerprinting technique.
6. How are stem cells useful in regenerative process?
7. Differentiate between outbreeding and inbreeding.

VIII Long answers questions

1. What are the effects of hybrid vigour in animals.
2. Describe mutation breeding with an example.
3. Biofortification may help in removing hidden hunger. How?
4. With a neat labelled diagram explain the techniques involved in gene cloning.
5. Discuss the importance of biotechnology in the field of medicine.

IX Higher Order Thinking Skills (HOTS)

1. A breeder wishes to incorporate desirable characters into the crop plants. Prepare a list of characters he will incorporate
2. Organic farming is better than Green Revolution. Give reasons
3. Polyploids are characterised by gigantism. Justify your answer.
4. 'P' is a gene required for the synthesis of vitamin A. It is integrated with genome of 'Q' to produce genetically modified plant 'R'.

   i. What is P, Q and R?

   ii. State the importance of 'R' in India.

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